

Convolution

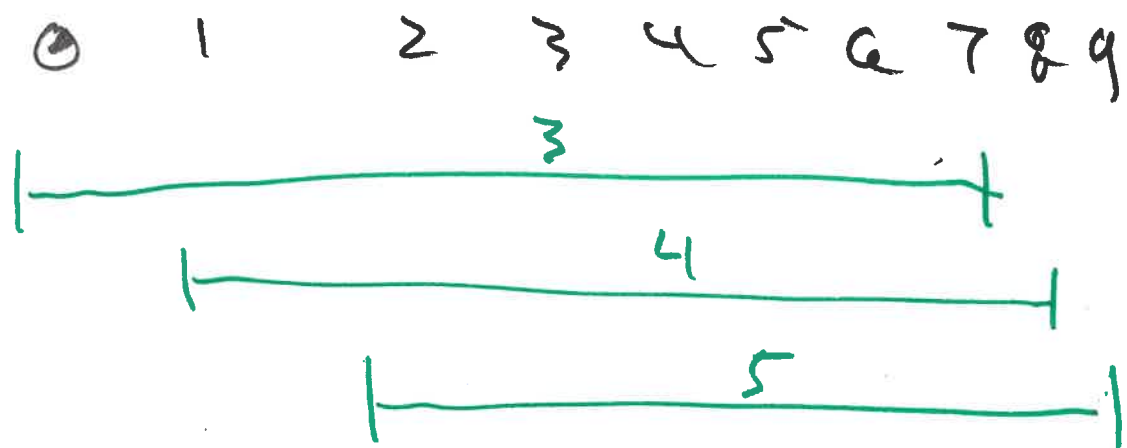
$$F * g(x) = \int_{-\infty}^{\infty} \cancel{F(x-y)} g(y) dy$$

$$F * g = a * b_j = \sum_{i=1}^n a[i] b[j-i]$$

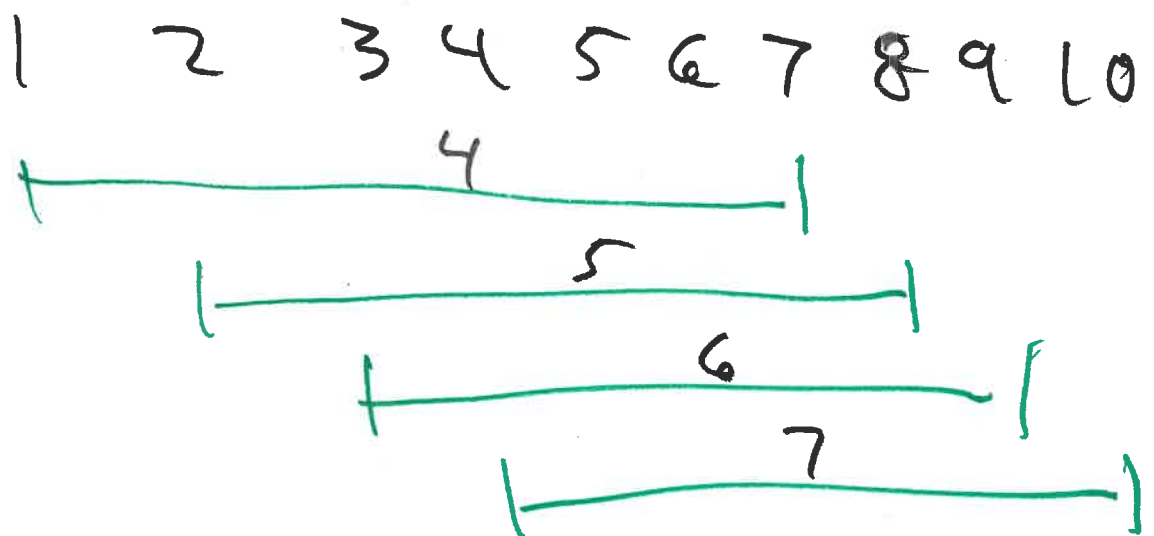
$$\underbrace{b[i-1] + b[i] + b[i+1]}_3$$

$$\frac{1}{3} b[i-1] + \frac{1}{3} b[i] + \frac{1}{3} b[i+1]$$

$$a = \left[\frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right]$$



1 4, 5, 6 10



B 10×10

U 10×10 S 10×10 V^t 10×10

U_{22} 10×2 S_{22} 2×2 V^t 2×10

$= 10 \times 10$ Rank = 2